2.10.51

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Question:If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are three non-zero, non-coplanar vectors and

$$\mathbf{b}_1 = \mathbf{b} - \frac{\mathbf{b}\cdot\mathbf{a}}{|\mathbf{a}|^2}\mathbf{a}, \qquad \mathbf{b}_2 = \mathbf{b} + \frac{\mathbf{b}\cdot\mathbf{a}}{|\mathbf{a}|^2}\mathbf{a},$$

$$\mathbf{c}_1 = \mathbf{c} - \frac{\mathbf{c}\cdot\mathbf{a}}{|\mathbf{a}|^2}\mathbf{a} + \frac{\mathbf{b}\cdot\mathbf{c}}{|\mathbf{c}|^2}\mathbf{b}_1, \mathbf{c}_2 = \mathbf{c} - \frac{\mathbf{c}\cdot\mathbf{a}}{|\mathbf{a}|^2}\mathbf{a} + \frac{\mathbf{b}_1\cdot\mathbf{c}}{|\mathbf{b}_1|^2}\mathbf{b}_1,$$

$$\mathbf{c}_3 = \mathbf{c} - \frac{\mathbf{c}\cdot\mathbf{a}}{|\mathbf{c}|^2}\mathbf{a} + \frac{\mathbf{b}\cdot\mathbf{c}}{|\mathbf{c}|^2}\mathbf{b}_1, \mathbf{c}_4 = \mathbf{c} - \frac{\mathbf{c}\cdot\mathbf{a}}{|\mathbf{c}|^2}\mathbf{a} = \frac{\mathbf{b}\cdot\mathbf{c}}{|\mathbf{b}|^2}\mathbf{b}_1,$$

then the set of orthogonal vectors is

1) $(\mathbf{a}, \mathbf{b}_1, \mathbf{c}_3)$ 2) $(\mathbf{a}, \mathbf{b}_1, \mathbf{c}_2)$ 3) $(\mathbf{a}, \mathbf{b}_1, \mathbf{c}_1)$ 4) $(\mathbf{a}, \mathbf{b}_2, \mathbf{c}_2)$

solution: Let $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ be three linearly independent (non-coplanar) vectors in $\mathbb{R}^3$.
We form a 3×3 matrix whose columns are these vectors:

$$A = \begin{pmatrix} | & | & | \\ \mathbf{a} & \mathbf{b} & \mathbf{c} \\ | & | & | \end{pmatrix} \tag{1}$$

To obtain an orthogonal set, apply the projection formula (Gram-Schmidt):
Let $\mathbf{v}_1 = \mathbf{a}$. Orthogonalize $\mathbf{b}$ against $\mathbf{a}$:

$$\mathbf{v}_2 = \mathbf{b}_1 = \mathbf{b} - \text{proj}_{\mathbf{a}}\mathbf{b} = \mathbf{b} - \frac{\mathbf{a}^T\mathbf{b}}{\mathbf{a}^T\mathbf{a}}\mathbf{a} \tag{2}$$

Orthogonalize $\mathbf{c}$ against $\mathbf{a}$ and $\mathbf{b}_1$:

$$\mathbf{v}_3 = \mathbf{c}_1 = \mathbf{c} - \frac{\mathbf{a}^T\mathbf{c}}{\mathbf{a}^T\mathbf{a}}\mathbf{a} - \frac{\mathbf{b}_1^T\mathbf{c}}{\mathbf{b}_1^T\mathbf{b}_1}\mathbf{b}_1 \tag{3}$$

Thus, the matrix with orthogonal columns is:

$$Q = \begin{pmatrix} | & | & | \\ \mathbf{a} & \mathbf{b}_1 & \mathbf{c}_1 \\ | & | & | \end{pmatrix} \tag{4}$$

The columns in Q form an orthogonal set.The set of orthogonal vectors is: $(\mathbf{a}, \mathbf{b}_1, \mathbf{c}_1)$